ReactJS.

The Component model

Topics

- **Component State.**
  - **Basis for dynamic, interactive UI.**
- **Data Flow patterns.**
- **Hooks.**

Component Data

- A component has **two** sources of data:

  1. **Props** - Passed into a component; Immutable; the props object.
  2. **State** - Internal to the component; Causes the component to re-render when changed / mutated.

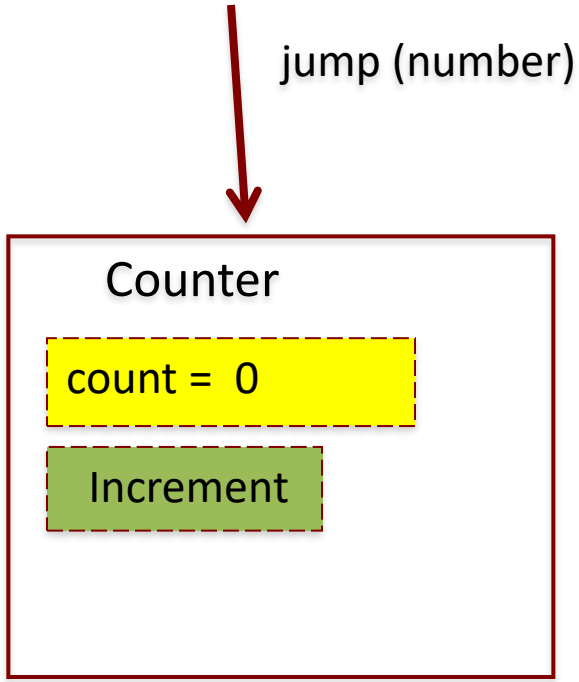
  – Both can be any data type - primitive, object, array.

- Props-related features:
  – Default values.
  – Type-checking.

- State-related features:
  – Initialisation.
  – Mutation – using a setter method.
    – Automatically causes component to re-render. ***
    – Performs an overwrite operation, not a merge.

Stateful Component Example

- **The Counter component.**
- **Ref: Examples Archive in this topic**
- **The** useState() **function:**
  - **Declares a state variable.**
  - **Returns a Setter / Mutator method.**
  - **Termed a React hook.**




```react/components/Counter.tsx
import { useState } from "react";

type CounterProps = {
  jump: number,
}

const Counter = (props: CounterProps) => {
  const [count, setCount] = useState(0);
  console.log(`${count} ${props.jump}`);

  const incrementCount = () => {
    setCount(count + props.jump)
  };

  return (
    <>
      <h2>Count: {count}</h2>
      <h3>Increment size: {props.jump}</h3>
      <button type="button" onClick={incrementCount}>
        Increment
      </button>
    </>
  );
};

export default Counter;
```

React’s event system.

- **Cross-browser support.**
- **Event handlers receive a React Event Object – a cross-browser wrapper for the browser's native event.**
- **React event naming convention slightly different to native:**

React	Native
onClick	onclick
onChange	onchange
onSubmit	onsubmit

- See https://react.dev/learn/responding-to-events and https://react.dev/reference/react-dom/components/common#react-event-object

for full details

Automatic Re-rendering.

- **EX.: The Counter component.**

User clicks button

→ onClick event handler executes (incrementCounter)

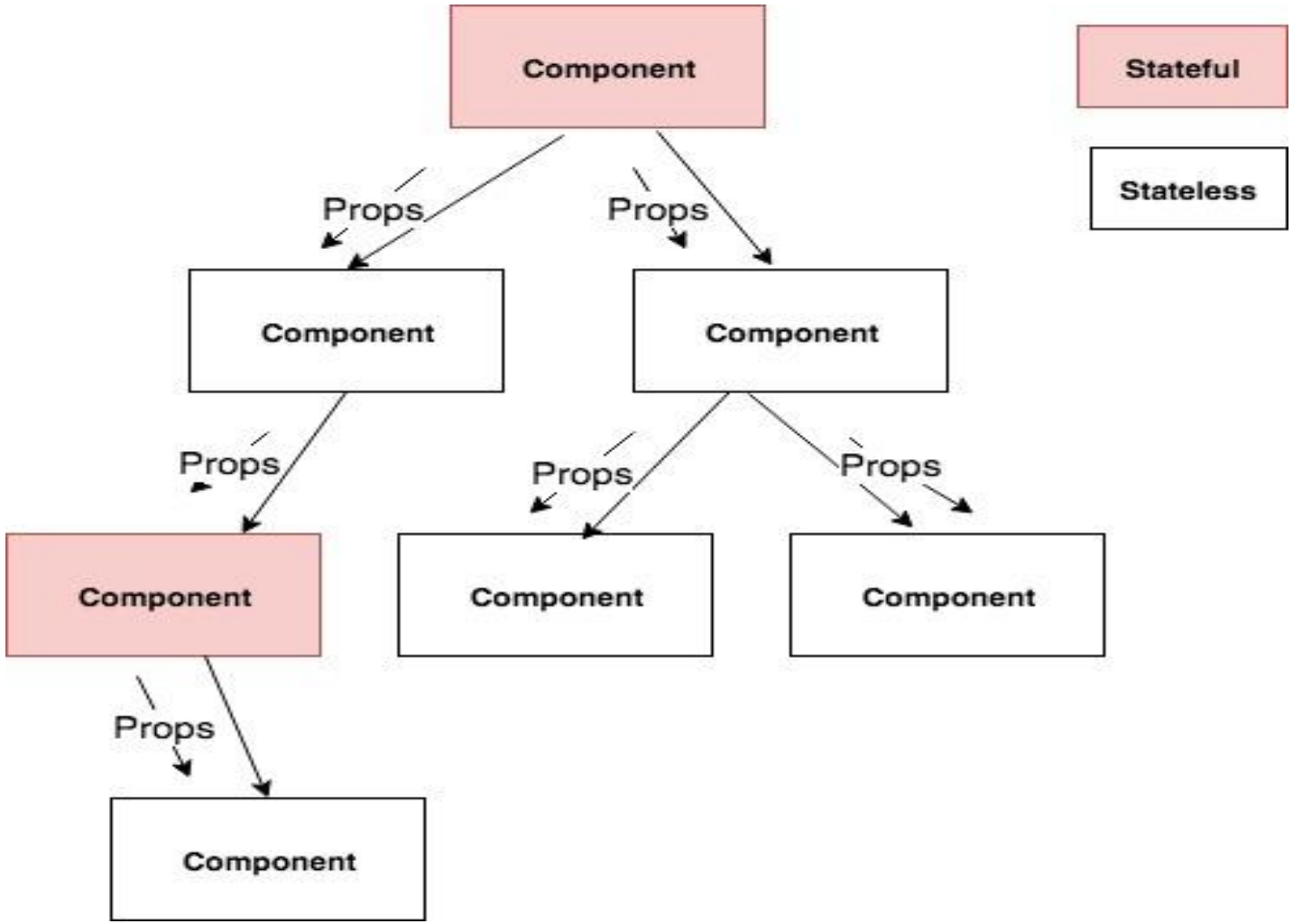
→ handler changes component's state (setCount())

*→ component function re- executed (**re-rendering**)*

Topics

- **Component State.** ✔
- **Data Flow patterns.**
- **Hooks.**

Unidirectional data flow



Unidirectional data flow

- In React, data flow is **unidirectional ONLY.**
  - Other SPA frameworks use two-way data binding.

- Typical React app pattern: **A small subset of the components are stateful**; the majority are stateless.
- Stateful component execution flow:
  1. User interaction causes a component state change.
  2. Component re-renders (re-executes).
  3. Component recomputes the props for its subordinate components.
  4. Subordinate components re-render, and recomputes props for its subordinates.
  5. etc.

Topics

- **Component State.** ✔
- **Data Flow patterns.** ✔ *(more later)*
- **Hooks**

React Hooks

- **A React Hook is a special function that lets you "hook into" React features (like state and lifecycle) inside functional components.**
- **React Hooks are:**
  1. **Functions (some are Higher Order Functions).**
  2. **That performs component** *state manipulation* **and** *lifecycle management*.

- **Examples: useState, useEffect, useContext, useRef, etc**
  - **'use' prefix is necessary for linting tools.**

- **Usage Rule:**
  - **Can only call hooks at the 'top level' in a component.**
    - **i.e. Don't call hooks inside loops or condition statements.**

useEffect Hook

- **Used when a component needs to perform** side effects**.**

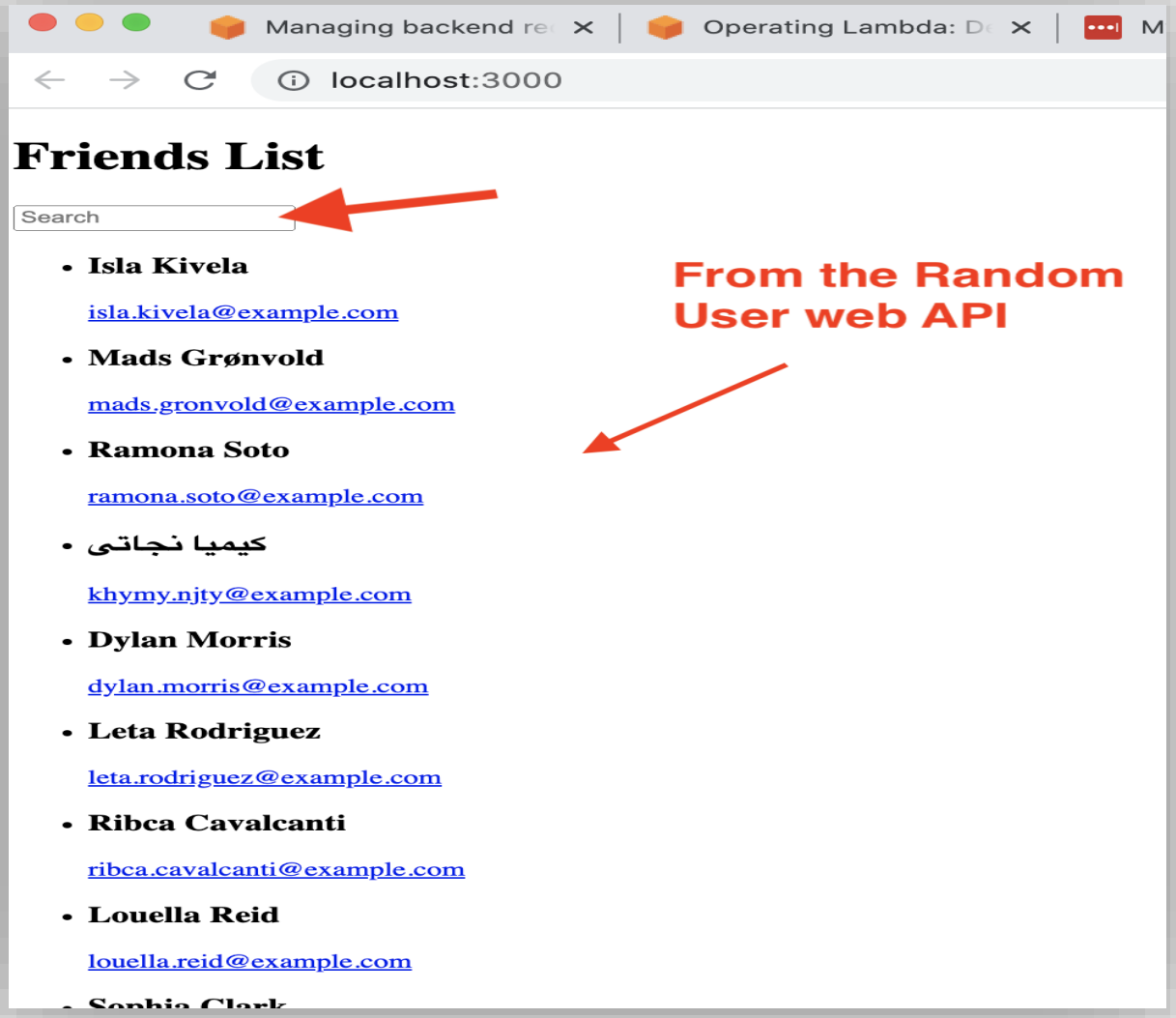
- **Side Effect example:**
  - **fetching data from a web API.**
  - **Subscribe to a browser events, e.g. window resize.**

- **Signature:** useEffect(callback, dependency array)
  - **The** *callback* **contains the side effect code.**
  - Dependencies **determine when a hook is executed.**

- **useEffect is executed:**
  1. **On component mounting.**
  2. **On every rendering that coincides with a dependency entry update.**

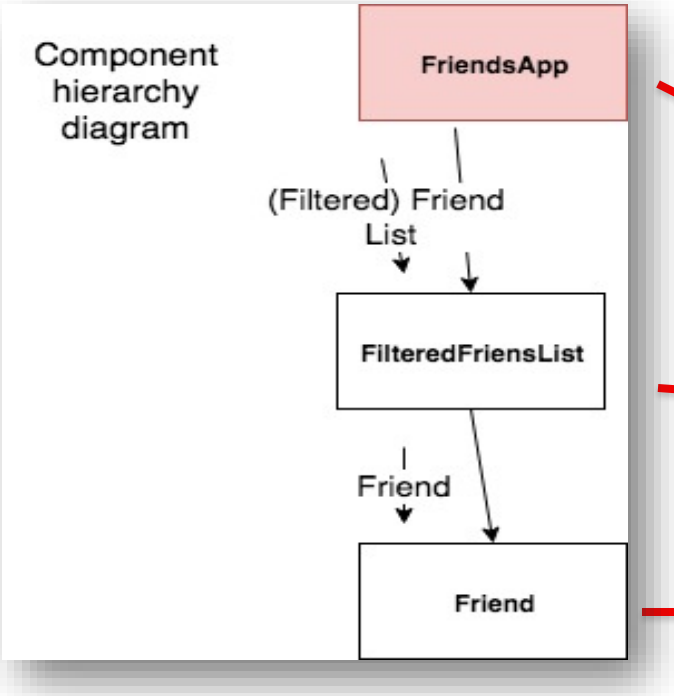
  - **An** empty dependency array **restricts execution to mount-time only.**

Sample App


Managing backend re
Operating Lambda: D
M
localhost:3000
Friends List
Search
Isla Kivela
isla.kivela@example.com
Mads Grønvold
mads.gronvold@example.com
Ramona Soto
ramona.soto@example.com
كيميا نجاتى
khymy.njty@example.com
Dylan Morris
dylan.morris@example.com
Leta Rodriguez
leta.rodriguez@example.com
Ribca Cavalcanti
ribca.cavalcanti@example.com
Louella Reid
louella.reid@example.com
From the Random User web API

Sample App 1

(see lecture archive)

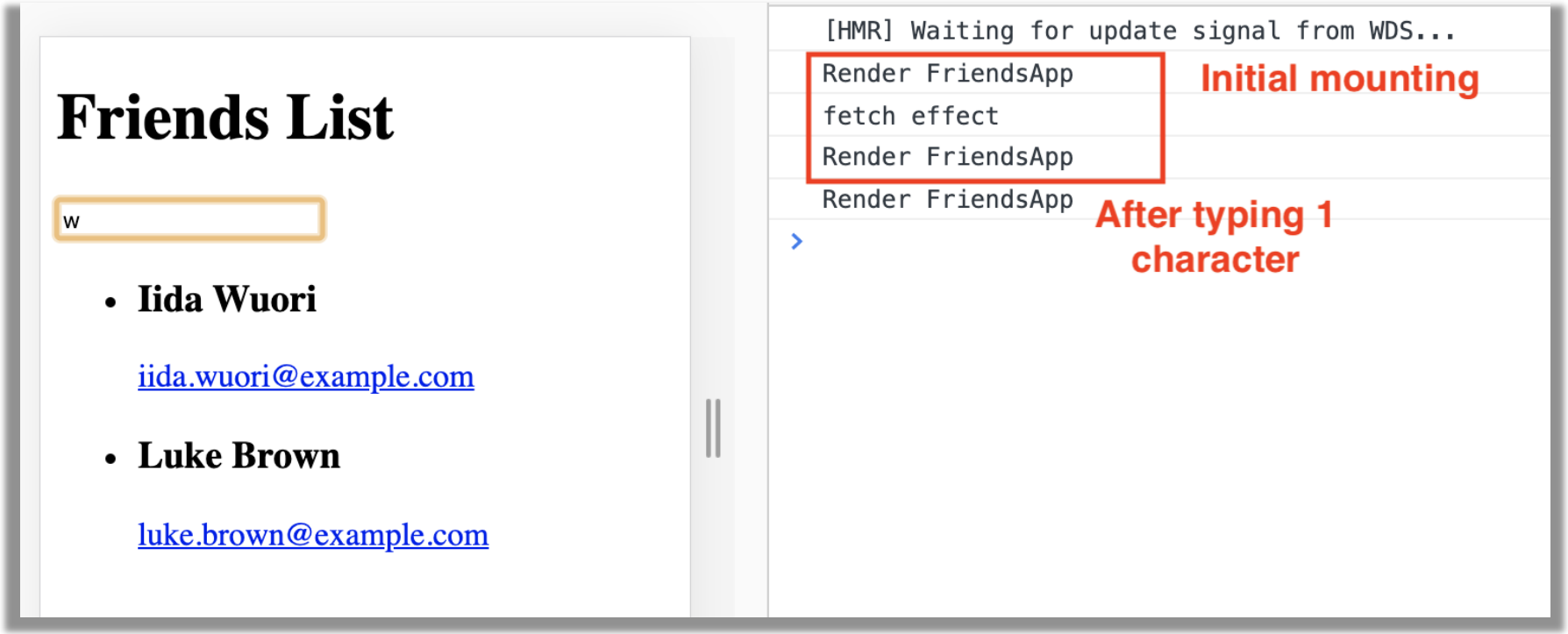


FriendsApp **component:**

1. **Manages** state **(i.e. text box, full list of friends).**
2. **Fetches data from a web API – side effect.**
3. **Computes matching friends.**
4. **Controls list rendering.**

Sample App - *useEffect* Hook

- **useEffect runs AT THE END of a component's mount process.**
  i.e. First rendering occurs BEFORE the API data is available.
  - **We must accommodate this in the implementation.**



Sample App - *useEffect* Hook

- **You must allow for asynchronous nature of API calls:**
  1. **UI must remain interactive while waiting for API data.**
  2. **Components should render without errors before API data is available.**

- **Correct approach:**

  const [friends, setFriends] = useState(***[ ]***) ; // Store API data

- **Incorrect approach:**

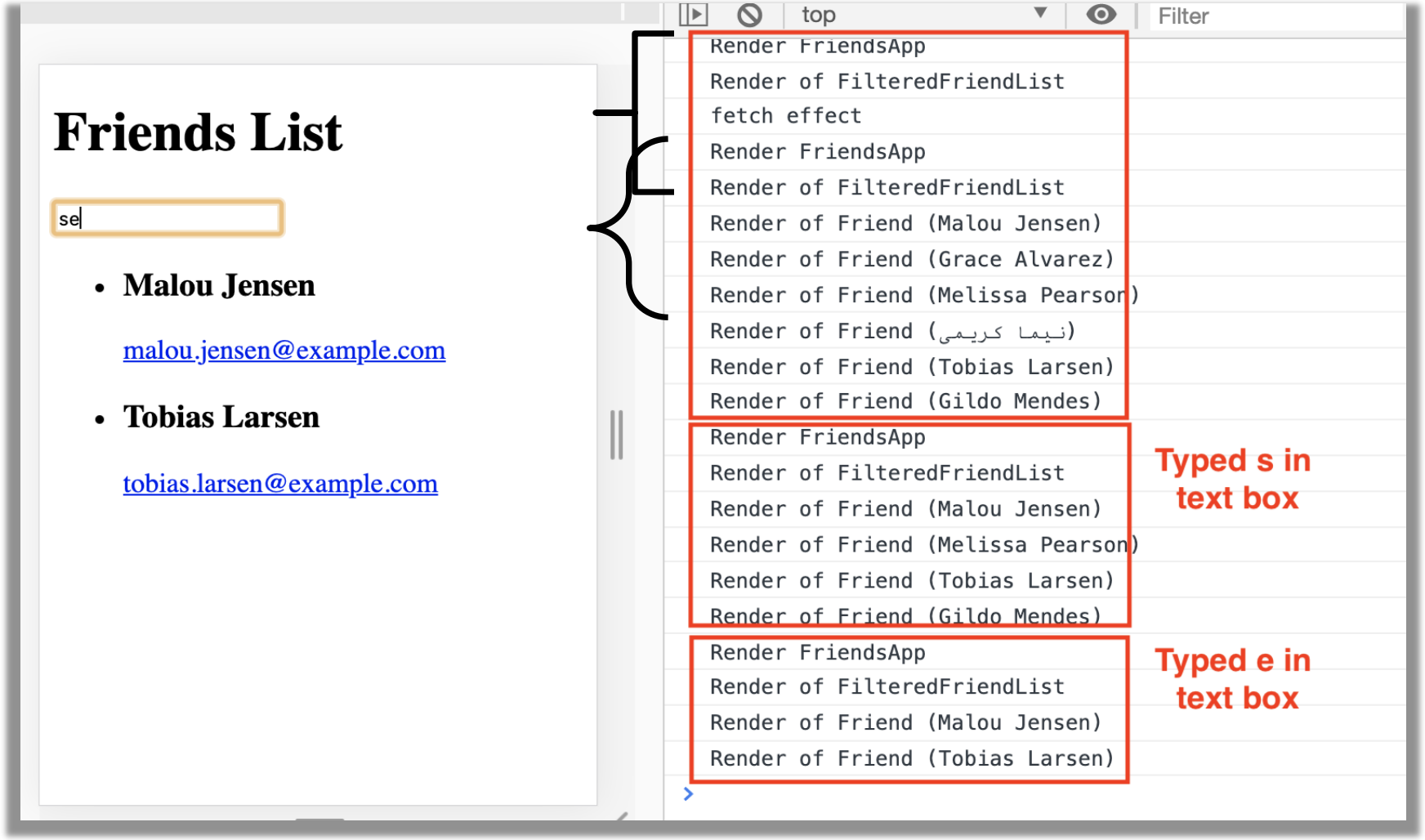
  const [friends, setFriends] = useState(***null***);

  – **Resulting error**

  TypeError: Cannot read property 'filter' of null

Unidirectional data flow & **Re-rendering**

(Assume we request 6 friends from web API)



More to come ….

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